

MPI Refresher

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An aside on parallelization

- **Shared** memory parallelism
 - **CUDA** (single GPU), **OpenMP**
- All threads run on the **same compute node** (the same GPU)
- All threads can access a **common memory**
- Data exchange via **shared memory** spaces
- **Distributed** memory parallelism
 - **MPI**, **PGAS**
- Processes can run on **different compute nodes**
- Each process works on **its own memory**
- Data exchange via **explicit communication** between ranks

An aside on parallelization

- MPI and OpenMP are standardized
 - MPI 4.1 (11/2023) – 1166 pages
 - OpenMP 5.2 (11/2021) – 669 pages
- Implementation is separated
 - Open MPI, MPICH, MVAPICH
 - Intel MPI, CRAY MPICH, IBM MPI, ...
 - OpenMP support in various compilers^[1]
 - Implemented specification version may vary
- Roots in CPU-only applications, extensions for GPUs

[1] <https://www.openmp.org/resources/openmp-compilers-tools/>

From CUDA to MPI

- CUDA
 - Each CUDA thread executes the kernel body
 - Threads can be identified by their thread and block IDs
 - Without synchronization, all ranks execute asynchronously
- MPI
 - Each MPI rank executes the whole application
 - Ranks can be identified by their rank IDs
 - Without synchronization, all ranks execute asynchronously

MPI Hello World

- Hello world example

```
#include <mpi.h>
```

```
int main(int argc, char *argv[]) {  
    MPI_Init(&argc, &argv);
```

```
    MPI_Finalize();  
}
```

MPI Hello World

- Hello world example

```
#include <mpi.h>

int main(int argc, char *argv[]) {
    MPI_Init(&argc, &argv);

    int rank, numRanks;
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &numRanks);

    MPI_Finalize();
}
```

MPI Hello World

- Hello world example

```
#include <mpi.h>

int main(int argc, char *argv[]) {
    MPI_Init(&argc, &argv);

    int rank, numRanks;
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &numRanks);

    std::cout << "Hello from rank " << rank << " of " << numRanks << std::endl;

    MPI_Finalize();
}
```

MPI Hello World

- Compile with

```
mpicxx -O3 mpi-hello-world.cpp \  
-o mpi-hello-world
```

- Run with

```
mpirun -n 4 ./mpi-hello-world
```

- Could print

```
Hello from rank 0 of 4  
Hello from rank 1 of 4  
Hello from rank 2 of 4  
Hello from rank 3 of 4
```

- But could also print

```
Hello from rank 2 of 4  
Hello from rank 3 of 4  
Hello from rank 1 of 4  
Hello from rank 0 of 4
```

Communication

- Communication between ranks via point-to-point and collective operations
- Point-to-point: send and receive messages between two ranks, e.g.

MPI_Send

MPI_Recv

- Collective: all ranks participate in the same operation, e.g.

MPI_Barrier

MPI_Reduce

MPI_Bcast

Example

Rank 1

Rank 3

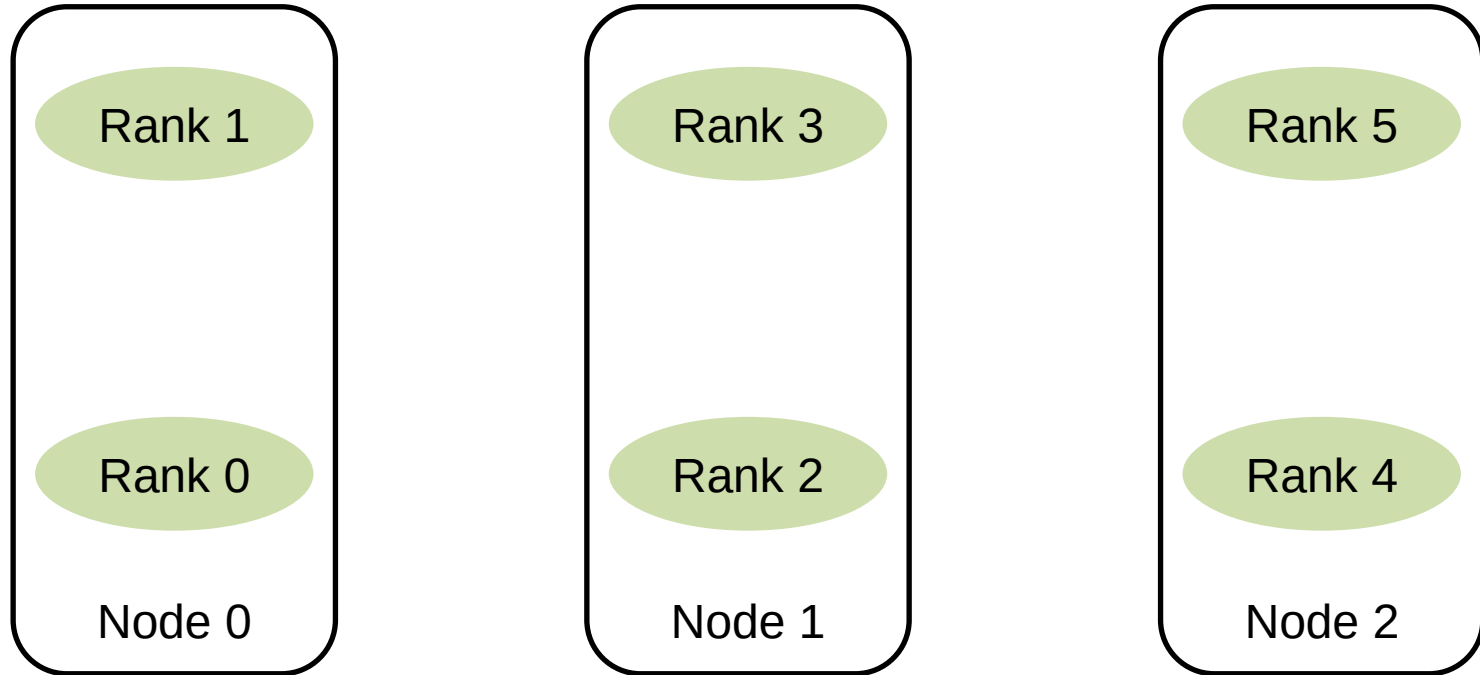
Rank 5

Rank 0

Rank 2

Rank 4

Example Node Mapping



Example with Values

Rank 1
5

Rank 3
0

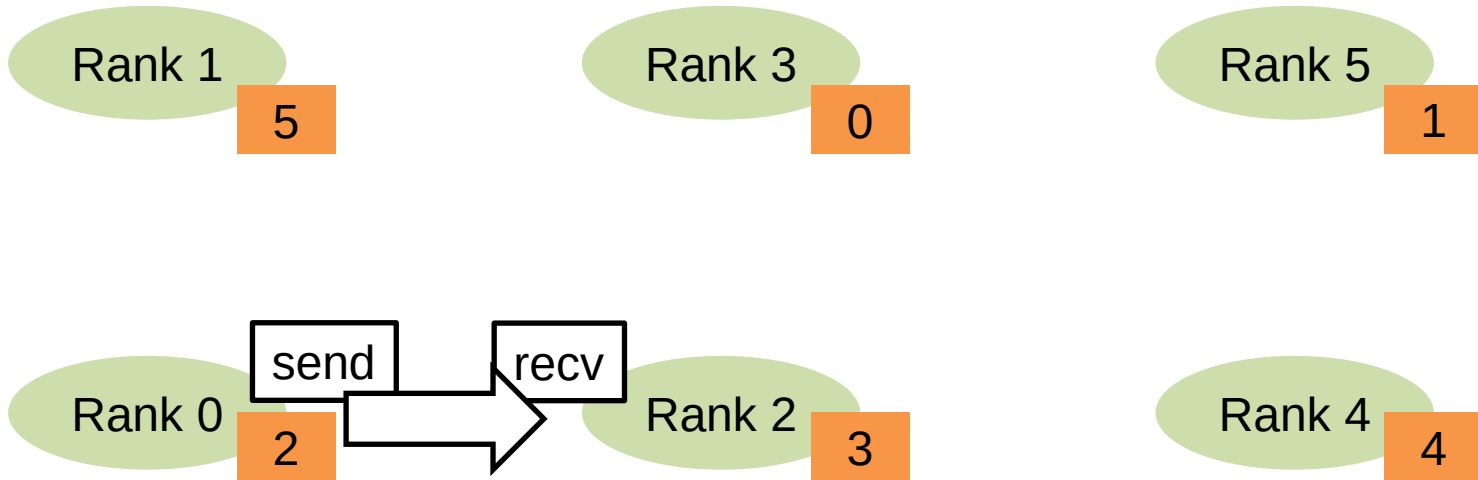
Rank 5
1

Rank 0
2

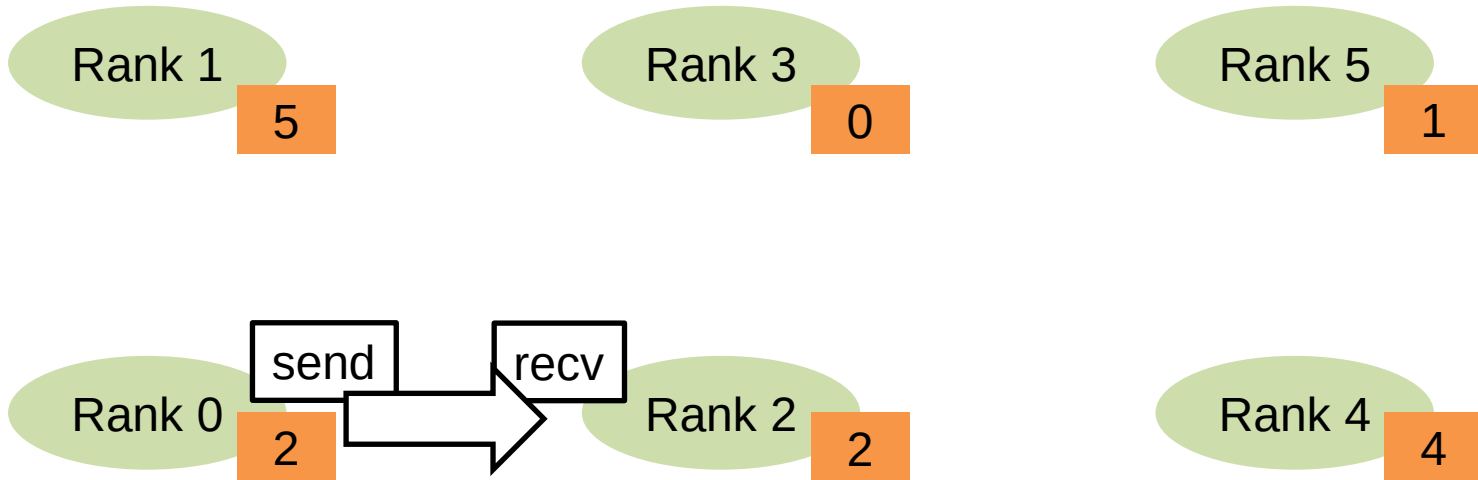
Rank 2
3

Rank 4
4

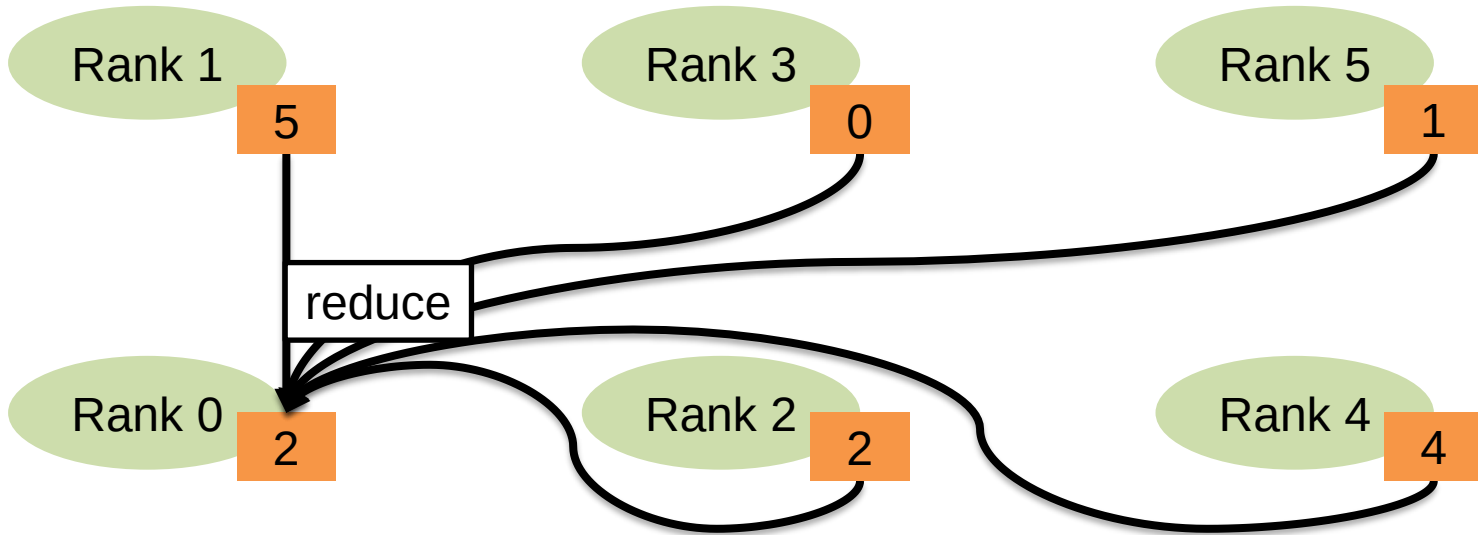
MPI Send/Recv



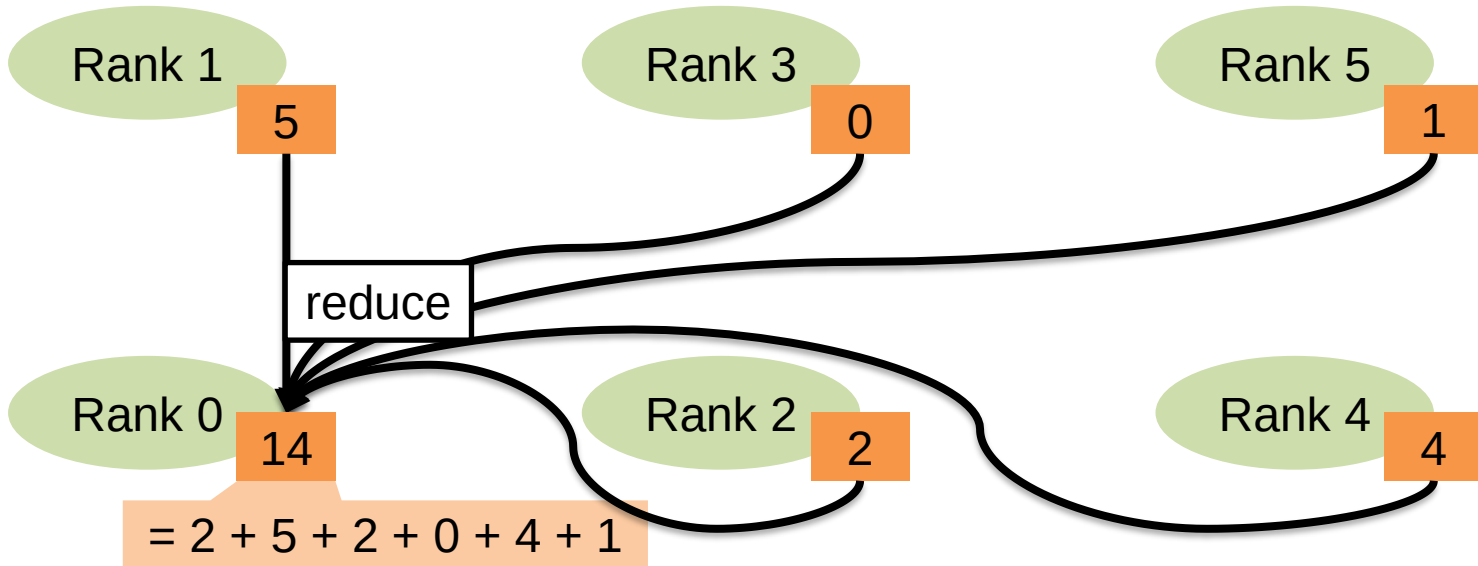
MPI Send/ Recv



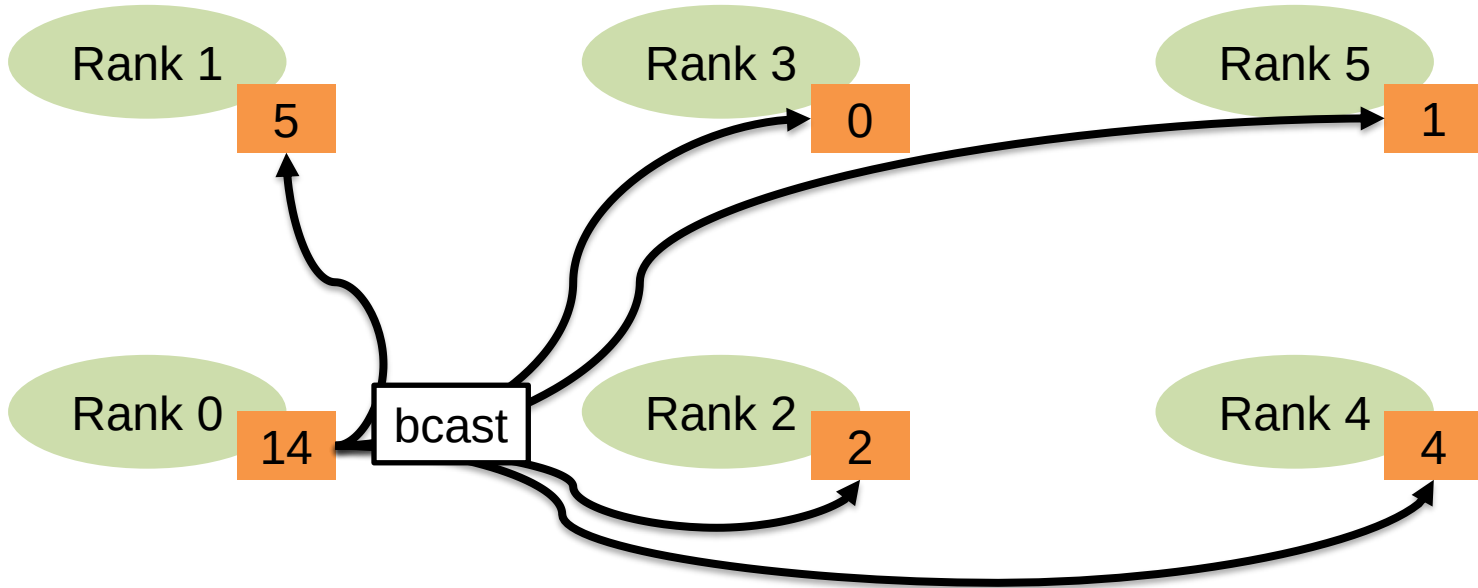
MPI Reduce



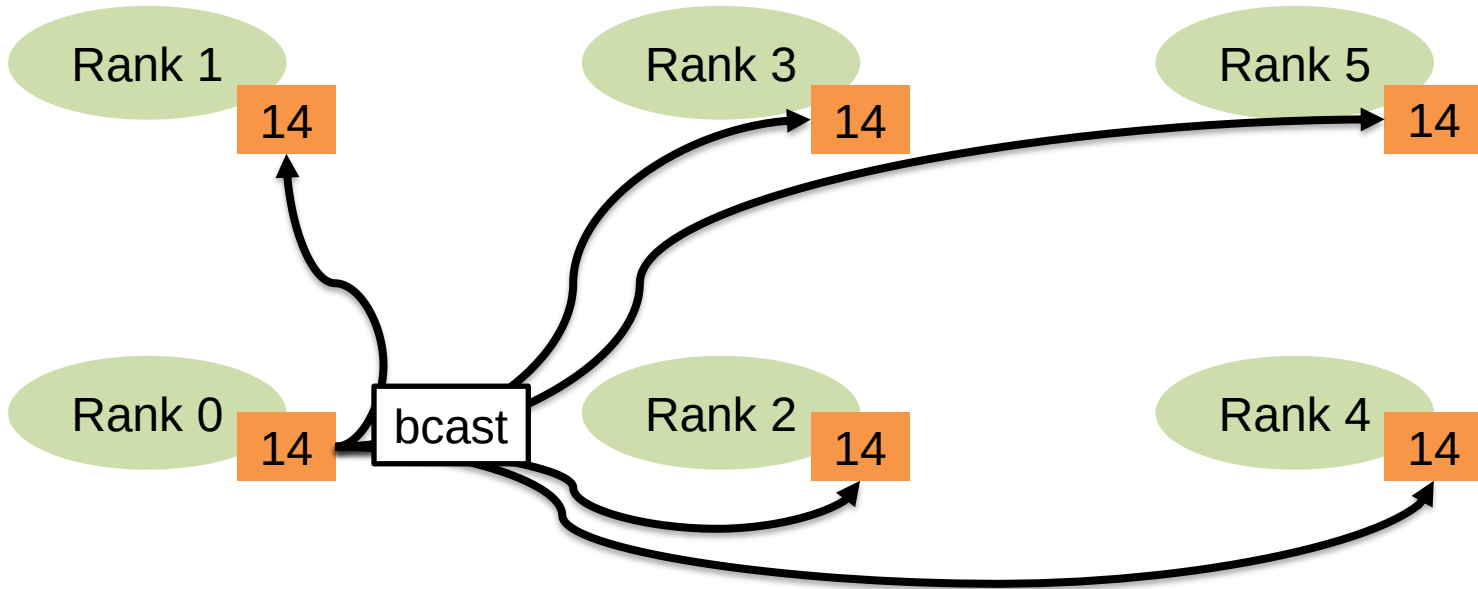
MPI Reduce



MPI Broadcast



MPI Broadcast



Communication

- Communication between ranks via point-to-point and collective operations
- Point-to-point: send and receive messages between two ranks, e.g.

```
MPI_Send(&sendData, count, MPI_DOUBLE, dest, tag, MPI_COMM_WORLD);  
MPI_Recv(&recvData, count, MPI_DOUBLE, src, tag, MPI_COMM_WORLD, &mpiStatus);
```

- Collective: all ranks participate in the same operation, e.g.

```
MPI_Barrier(MPI_COMM_WORLD);  
MPI_Reduce(&sendData, &recvData, count, MPI_DOUBLE, MPI_SUM, root, MPI_COMM_WORLD);  
MPI_Bcast(&data, count, MPI_DOUBLE, root, MPI_COMM_WORLD);
```

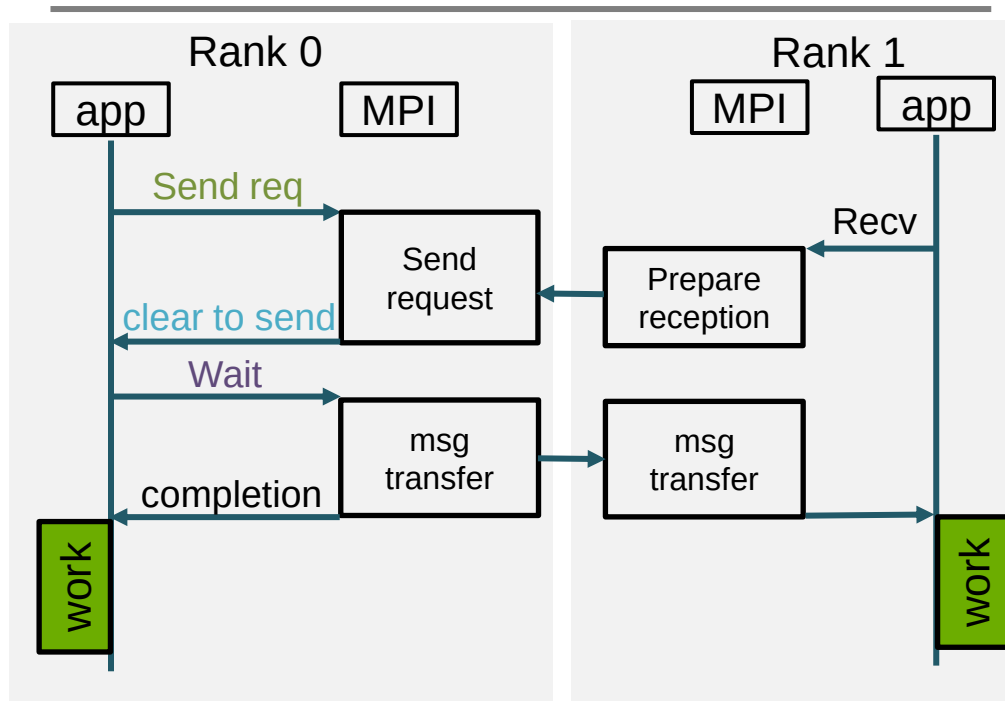
Beyond First Steps

- Other collective operations
 - MPI_Gather, MPI_Scatter, MPI_Scan, ...
 - MPI_Allreduce, MPI_Allgather, ...
- Non-blocking communication
 - MPI_Isend, MPI_Irecv, MPI_Ireduce, ...
 - MPI_Test, MPI_Wait, ...
- One-sided communication
 - MPI_Get, MPI_Put, ...
- Custom (sub-) communicators
 - MPI_Cart_create, MPI_Dims_create, ...
- MPI I/O, Error handling, timing/profiling, ...

CUDA-Aware MPI

MPI Synchronous communication

```
.  
.   
.   
if(rank == 0) {  
    MPI_Send(send_buffer, 1, MPI_INT,  
    rank+1, MPI_ANY_TAG, MPI_COMM_WORLD);  
} else if(rank == 1) {  
    MPI_Recv(recv_buffer, 1, MPI_INT,  
    rank-1, MPI_ANY_TAG, MPI_COMM_WORLD);  
}  
work();  
.   
.   
.
```



MPI Asynchronous communication

```
.  
.   
.   
if(rank == 0) {  
    MPI_Isend(send_buffer, 1, MPI_INT,  
    rank+1, MPI_ANY_TAG, MPI_COMM_WORLD);  
} else if(rank == 1) {  
    MPI_Recv(recv_buffer, 1, MPI_INT,  
    rank-1, MPI_ANY_TAG, MPI_COMM_WORLD);  
}  
work();  
.   
.   
.
```

Best case scenario

